RAG Assignment: Baseline vs. Re-Ranking

Analysis and Report

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Code: github.com/roshanned/rag_baseline_vs_reranking

1 Pipeline Setup

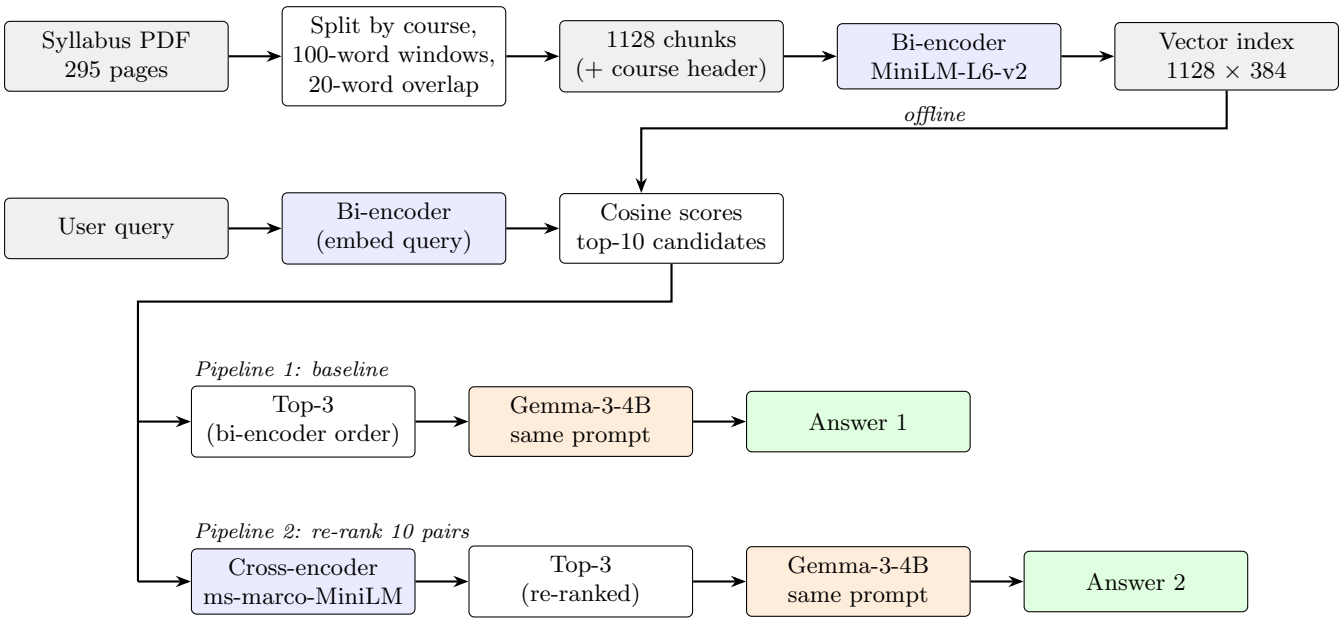


Figure 1: Setup

2 Architecture Comparison

- **Bi-encoder:** encodes the query q and chunk d *separately* into vectors $\mathbf{q}$, $\mathbf{d}$, compared by

$$\text{sim}(q, d) = \frac{\mathbf{q} \cdot \mathbf{d}}{\|\mathbf{q}\| \, \|\mathbf{d}\|}.$$

- Chunk vectors are precomputed once, so search is fast.
- **Cross-encoder:** reads (q, d) together as one input. Joint self-attention across all tokens gives one relevance score.
- More precise, but nothing can be precomputed. One forward pass per (query, chunk) pair at query time.

3 Qualitative and Rank Shift Analysis

3.1 Overview

I tested 31 queries covering keyword traps, exceptions, negation and buried facts. For each one I marked the gold chunk using an exact phrase from the PDF. Out of the 31, 22 already had the gold chunk in the baseline top 3, 2 were fixed by re-ranking, 7 were recall failures where the right chunk was not in the top 10, and none got worse after re-ranking.

Query	Rank of gold chunk (bi-encoder to cross-encoder)	Answer 1 (baseline)	Answer 2 (re-ranked)
Which course uses an open book exam followed by a project with flight testing?	UAV Design chunk (page 153): rank 4 to rank 1	"The provided passages do not contain the answer."	Correct, cited [1]: "Open book exam followed by detailed project submission including simulation studies, prototype development and flight testing."
Is there a limit on how many students can register for Autonomous Software Engineering?	Chunk on page 31: rank 5 to rank 1	Refused	Also refused
Which course has an L-T-P-C of 3-0-2-6? (recall failure)	Gold chunk at rank 424, never in the top 10	Refused	Refused

Table 1: Rank changes and answers for the three analysed queries.

3.2 Results by failure type

Failure type	Queries	Already fine	Fixed	Recall failure
Keyword trap	8	6	0	2
Exception	7	6	1	0
Negation	11	8	0	3
Buried fact	5	2	1	2
Total	31	22	2	7

Table 2: Outcome per failure type (gold chunk in the top 3 already, fixed by re-ranking, or outside the top 10).

Both fixes came from queries where the gold chunk sat at rank 4–10 and was pushed up by the cross-encoder: one keyword-trap/buried query (UAV) and one exception query (intake). All 7 recall failures had the gold chunk outside the top 10, so re-ranking could not help. Negation was the weakest type, with 3 of 11 queries never retrieved. No query got worse.

3.3 Detailed examples

Keyword trap + buried chunk: open book exam with flight testing. The query was "Which course uses an open book exam followed by a project with flight testing?"

- The baseline top 3 were Digital VLSI Design (page 83), Wireless Communications (page 295) and Probability and Statistics (page 210). They share words like "project" and "exam" with the question, so they are lexical distractors, but they are the wrong courses. Their bi-encoder scores (0.516, 0.511, 0.508) were almost the same as the right chunk's (0.507), which is why it was ranked 4th.
- The right chunk, from Introduction to UAV Design on page 153, moved from rank 4 to rank 1. Digital VLSI dropped from rank 1 to 7 and Probability and Statistics from 3 to 5.
- Answer 1 was "The provided passages do not contain the answer." since the right chunk was outside the top 3 and the model had no evidence.
- Answer 2 quoted the correct sentence: "Open book exam followed by detailed project submission including simulation studies, prototype development and flight testing."

Exception / corner case: intake limit of Autonomous Software Engineering. The query was "Is there a limit on how many students can register for Autonomous Software Engineering?"

- The baseline top 3 were all general chunks about the project and assessment (page 33). The specific field "In Take Limit : 30" sits in the course header chunk on page 31, which was at rank 5. The cross-encoder moved it to rank 1, and chunk 127 dropped from rank 3 to 4.
- Both Answer 1 and Answer 2 were "The provided passages do not contain the answer." So re-ranking fixed the retrieval but not the answer. The chunk only has the bare field "In Take Limit : 30", while the question asks how many students can register.
- When I asked "What is the In Take Limit of Autonomous Software Engineering?", the model answered "The In Take Limit of Autonomous Software Engineering is 30 [1]." So the problem was the difference in wording and the strict refusal rule in the prompt, not retrieval.

Negation and constraints.

- 8 of the 11 negation queries were already fine. For example, "Which prerequisite course is mandatory, not just desired, for Autonomous Software Engineering?" had the right chunk at rank 2 and re-ranking moved it to rank 1.
- The other 3 were recall failures: "Which course has no mid-semester exam?" (rank 475), "Which course forbids using ChatGPT for written work unless the faculty allows it?" (rank 221) and "Which course replaces the end-semester exam with a mini-project demo?" (rank 22).
- These questions describe a missing or forbidden thing without naming a course. The bi-encoder embeds "no mid-sem exam" close to every chunk that talks about mid-sem exams, so the real chunk is lost among hundreds of similar ones. Since it is outside the top 10, the re-ranker never sees it.

Recall failures in general.

- In all 7 recall failures the right chunk was outside the top 10 (ranks 13 to 475). For the L-T-P-C query it was at rank 424, so both pipelines refused.
- A re-ranker can only reorder the candidates it is given, so it cannot fix this.
- Exact tokens like 3-0-2-6 are also hard for a semantic embedding to match.

Without retrieval.

- When I asked the model the same questions with no passages, it answered confidently but wrongly, for example by inventing Udacity details and naming "Aviation Meteorology".
- This is why the model needs retrieval for a document it has never seen.

4 Latency Profiling

I ran each query 10 times on a T4 GPU, skipped the slow first runs, made sure the clock only stopped after the GPU finished, and averaged everything over 31 queries on a collection of 1128 chunks.

Stage	Mean time (ms)
Pipeline 1: embed query + score all chunks + top-10	9.6
Re-ranking step only: cross-encoder on 10 pairs	31.9
Pipeline 2: baseline retrieval + re-ranking	43.2
LLM answer generation (same for both)	2714.4

Table 3: Latency per query.

Re-ranking makes retrieval about $43.2/9.6 \approx 4.5\times$ slower, but this adds only about 34 ms, small next to roughly 2.7 s of LLM generation. (The LLM time was measured on 3 queries, so it is approximate.)

5 Scaling Question

The bi-encoder prepares the vector of every chunk once, in advance. When a question comes, it only converts the question into a vector and compares it with the saved ones, which is fast.

The cross-encoder cannot prepare anything in advance, because its score depends on the question and the chunk together. So for every question it has to read every chunk again from scratch. Its time therefore grows with the number of chunks N:

$$T_{\text{cross}} \approx N \cdot t_{\text{pair}},$$

where $t_{\text{pair}} \approx 2.82$ ms was measured in my run.

Approach	Time per query
Bi-encoder over 1128 chunks	9.6 ms
Cross-encoder on top 10 only	31.9 ms
Cross-encoder on all 1128 chunks	3.18 s
Cross-encoder on 10^6 chunks (extrapolated)	$\approx$ 47 min

Table 4: Cost of cross-encoding the whole collection. The last row is a straight-line estimate from the T4 measurement.

Even on this small collection, running the cross-encoder on everything is about $3180.6/9.6 \approx 331$ times slower than the bi-encoder. For a million chunks, one question would take about 47 minutes, which is not usable.